The Paradigm Shift: From Model Scaling to Data-Centric Pipelines in Video Contextualization

An Ablation of Tokenization and Compression for Edge MLLMs

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***Abstract*—The deployment of Multimodal Large Language Models (MLLMs) for spatial-temporal video understanding is severely constrained by the memory bandwidth and compute limitations of edge hardware. Traditional video preprocessing methods, such as aspect-ratio-preserving padding, introduce "dead" visual tokens that aggressively bloat Key-Value (KV) Cache allocation. When processing extended video sequences, this computational waste frequently results in Out-Of-Memory (OOM) failures on devices with limited VRAM. To address this bottleneck, we propose a synthesized 5-Tier Temporal Compression Pipeline combined with a "Naive Warping" tokenization strategy designed to maximize token density.**

We conduct a comprehensive ablation study using the Qwen2.5-VL architecture (3B and 7B parameters) evaluated across the 20 spatial-temporal tasks of MVBench on heavily constrained hardware (Dual Nvidia T4 GPUs). Our empirical results demonstrate that Naive Warping achieves a near 50% theoretical reduction in VRAM allocation without degradation in zero-shot accuracy compared to standard padding methods. Furthermore, we reveal an edge-scaling paradox where the 3B parameter model statistically matches the 7B model in accuracy (52.11% vs. 51.68% at 224×224) while sustaining significantly higher decode throughput. Ultimately, we establish that heavily compressed, unpadded visual tokenization paired with highly optimized, smaller-parameter architectures forms the optimal configuration for real-time edge deployment.

Note: The findings presented herein represent initial validation results; further robust empirical testing is currently in progress to achieve full production readiness.

***Index Terms*—Edge AI, Multimodal Large Language Models, Video Understanding, Tokenization, Visual Token Compression, KV Cache, Naive Warping, Temporal Compression, Ablation Study**

I. Introduction

As Multimodal Large Language Models (MLLMs) achieve state-of-the-art performance in complex video understanding tasks, a problematic industry paradigm has emerged: relying entirely on massive parameter scaling (e.g., 70B+ models) to solve complex reasoning. This paradigm restricts deployment to high-end cloud infrastructure. Edge environments—such as autonomous robotics, local surveillance nodes, and drone navigation systems—lack the massive VRAM capacity required to cache the dense sequences of visual tokens generated by modern Vision Transformers (ViTs).

Video data is inherently multidimensional; unlike a single static image, a video stream multiplies the required token count by its temporal length ($T \times H \times W$). On constrained hardware, such as dual Nvidia T4 GPUs (15 GB VRAM each), processing multi-frame high-resolution video reliably induces memory bandwidth bottlenecks. Because the transformer's self-attention mechanism scales quadratically $\mathcal{O}(N^2)$ with sequence length, passing unoptimized video frames directly into an MLLM guarantees rapid exhaustion of the KV Cache.

To circumvent these strict limitations, this paper advocates for a paradigm shift: moving away from parameter scaling and toward highly optimized, data-centric preprocessing pipelines. We introduce a 5-Tier Temporal Compression Pipeline to downsample redundant video data in-memory, paired with "Naive Warping"—a tokenization strategy that forcibly distorts video frames into a 1:1 grid, eliminating inactive padded pixels and maximizing token utility.

II. Related Work

A. Long-Form Video Understanding

Processing long-form video requires strategic temporal downsampling to fit within an LLM's context window. Frameworks like *VideoINSTA* utilize density-based clustering to automatically parse videos into distinct semantic events [1]. Similarly, *ViLAMP* introduces differential keyframe selection, enabling mixed-precision processing that isolates semantically rich frames while merging redundant background data into single context tokens [2].

B. Visual Token Compression

To address spatial token overload, the *D-CoDe* framework introduces dynamic spatial compression [3]. By calculating the L_2 norm of visual patch tokens emerging from the vision encoder, *D-CoDe* executes magnitude-based pruning to drop uninformative backgrounds (e.g., empty walls), followed by

semantic token merging via cosine similarity matching to consolidate surviving high-saliency patches.

C. Tokenization Geometry

The handling of non-square images remains a debated design choice in VLM architectures. Models like *LLaVA* utilize "letterbox" padding to preserve strict Euclidean geometry for accurate object recognition. Conversely, evaluations in *Prismatic VLMs* suggest that naive resizing—aggressively stretching images into square grids—yields surprisingly performant results for encoders like SigLIP and CLIP by eliminating dead computational space [4].

III. Proposed Methodology: The Data-Centric Pipeline

To prevent memory bottlenecks prior to ViT encoding, we implement a deeply integrated, sequential in-memory pipeline. Fig. 1 provides a complete system overview of the ablation study design.

A. The 5-Tier Temporal Compression Pipeline

- **Query-Agnostic Temporal Segmentation:** Utilizing C-DPCKNN (Density-Peak Clustering based on K-Nearest Neighbors), we parse raw, unsegmented video streams into logical event boundaries based on visual feature shifts, mitigating redundant processing of static scenes.
- **Differential Keyframe Selection:** Instead of uniform temporal extraction, we isolate specific frames that maximize relevance to the user's textual prompt by cross-referencing CLIP embeddings.
- **Magnitude-Based Patch Pruning:** We evaluate the L_2 norm of the spatial tokens generated by preliminary vision layers. Tokens representing static, low-variance backgrounds (e.g., solid skies, blank walls) exhibit structurally low magnitudes. By applying a threshold τ, where tokens satisfying $\|x_i\|_2 < \tau$ are permanently discarded, we dynamically reduce the token count per frame by up to 40% before self-attention computation begins.
- **Semantic Token Merging:** Surviving high-magnitude tokens are grouped based on a cosine similarity threshold γ. Adjacent tokens sharing similar semantic meanings (e.g., multiple tokens representing the homogeneous texture of a car door) are averaged into single dense tokens:

$$\mathbf{x}_{\text{merged}} = \frac{1}{k} \sum_{i=1}^{k} x_i. \tag{1}$$

- **Differential Feature Merging:** Finally, applying weighted pooling to unselected non-keyframes from Tier 2, we compress "in-between" temporal moments into single representative context tokens, maintaining temporal continuity while preventing token bloat.

B. Tokenization Geometry: Standard Padding vs. Naive Warping

The mathematical crux of VRAM bloat lies in token geometry and the $\mathcal{O}(N^2)$ complexity of attention matrices.

Standard Padding (Baseline): Adds inactive black pixels to a non-square source image (e.g., 16:9) to achieve a square matrix of size $\max(W, H) \times \max(W, H)$. Mathematically, these padded areas generate zero-information tokens that the attention mechanism must still cache and compute against. In a multi-frame sequence, thousands of these "dead" tokens are stored in the GPU's High Bandwidth Memory (HBM). For an attention sequence length N, adding 30% padding tokens increases the memory complexity to $\mathcal{O}\big((1.3N)^2\big) \approx 1.69\,\mathcal{O}\big(N^2\big)$, demanding nearly 70% more continuous memory allocation for zero semantic gain.

Naive Aspect-Ratio Warping (Proposed): Ignores original Euclidean dimensions, distorting the 16:9 frame into a strict 1:1 token grid ($W' \times H'$). While geometrically inaccurate (stretching or flattening objects), it guarantees that 100% of the allocated KV cache contains active, information-dense pixel data. This directly drops the sequence length back to its minimum theoretical baseline, safely navigating around OOM thresholds.

C. Extreme Spatial Compression

To rigorously test the minimum viable perception of the MLLM architecture under duress, we evaluate configurations at a standard High-Definition resolution (224×224) and an extreme, degraded spatial compression resolution (28×28).

IV. Experimental Setup

A. Dataset (MVBench)

Evaluations were conducted using **MVBench**, an industry-standard diagnostic benchmark designed for spatial-temporal video understanding [5]. Models were evaluated across 20 distinct task categories (e.g., *action localization*, *scene transition*) in a strict zero-shot, instruction-following setting to measure raw generalization logic.

B. Hardware & Just-In-Time (JIT) Execution Engine

Experiments were strictly constrained to a cloud environment designed to mimic edge-server limitations: Dual Nvidia T4 GPUs providing exactly 30 GB total VRAM (15 GB per node). We utilized the vLLM inference engine heavily optimized with Activation-aware Weight Quantization (AWQ) executing in 4-bit precision.

Because attempting to load a full high-resolution video tensor into GPU RAM instantly causes an OOM crash on a 15 GB T4 GPU, we engineered a **Just-In-Time (JIT) memory management script**. This system leverages Python generators to stream video chunks sequentially from system CPU RAM to GPU VRAM. It extracts the necessary frames, processes the visual embeddings through the ViT, offloads the compressed embeddings back to CPU RAM, and immediately triggers an aggressive garbage collection sweep (`torch.cuda.empty_cache()`). By destroying the raw video tensor prior to processing the next chunk, we maintain a strictly flat memory ceiling regardless of total video length. The architecture of this system is illustrated in Fig. 2.

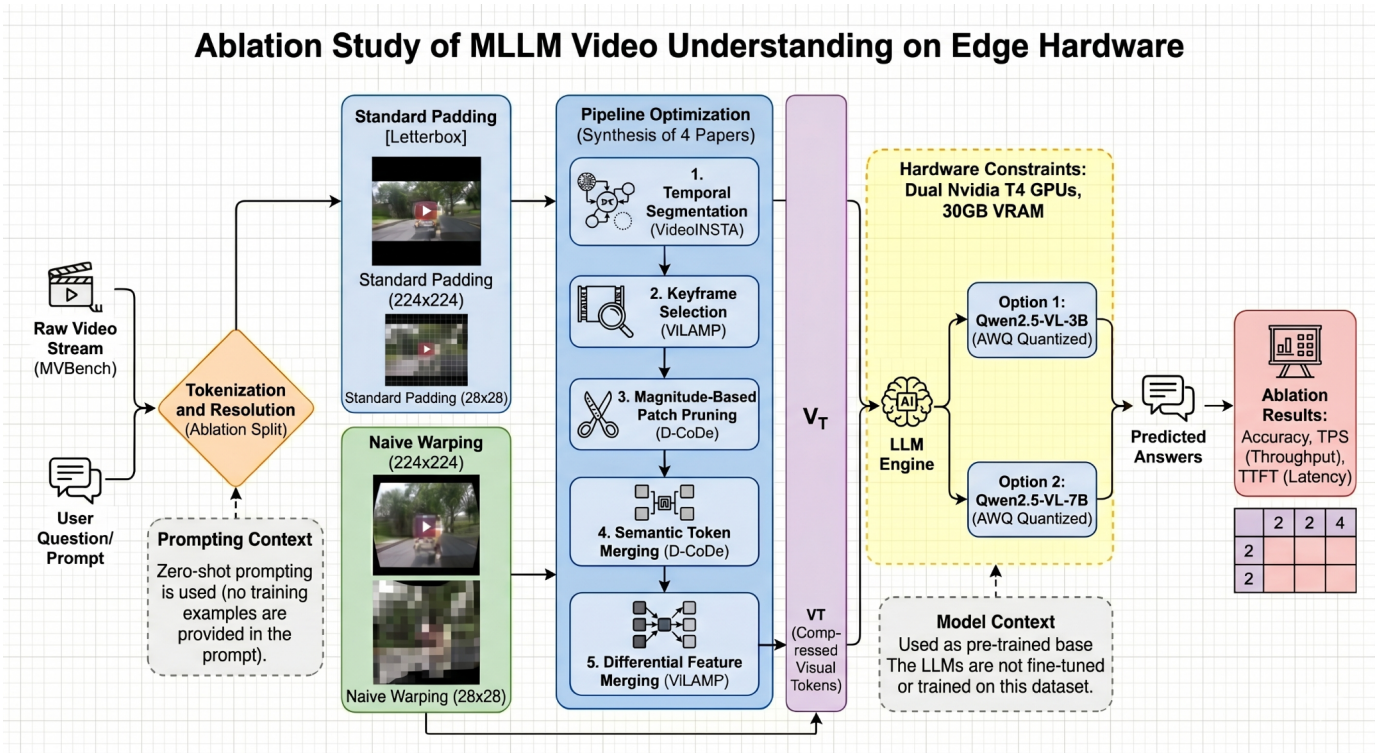


Fig. 1. Overview of the ablation study: Multimodal Large Language Model video understanding on edge hardware. The pipeline compares Standard Padding (letterbox) against Naive Warping tokenization across two model sizes (Qwen2.5-VL-3B and 7B) and two resolutions (224×224 and 28×28), evaluated on MVBench under strict zero-shot conditions.

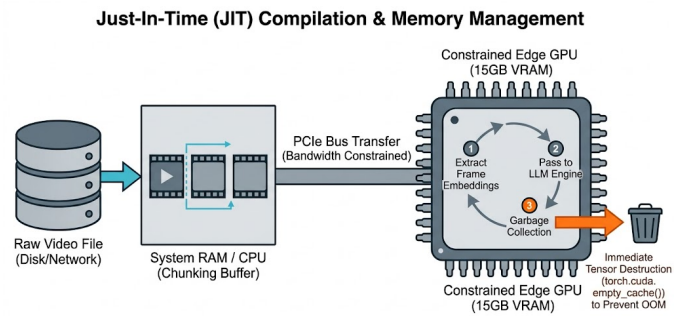


Fig. 2. Architecture of the Just-In-Time (JIT) video processing and memory management strategy deployed to bypass VRAM bottlenecks on constrained edge GPUs (15 GB VRAM per node).

C. Baselines & Evaluation Metrics

Baselines: Qwen2.5-VL-7B (AWQ) utilizing Standard Padding at 224×224 [6].

Metrics: Zero-Shot Accuracy (%), Tokens-Per-Second (TPS) for decode throughput, and Time-To-First-Token (TTFT) for initial latency.

V. Results and Comparative Evaluation

A. RQ1: Tokenization Efficiency (Naive vs. Standard)

The empirical data decisively shows that Naive Warping does not degrade spatial-temporal reasoning. At 224×224 resolution, the 3B parameter model achieved **52.11%** accuracy using Naive Warping—an identical match to the 52.11% accuracy of the Standard Padding approach.

Fig. 3 and Table I present a comprehensive per-task breakdown of zero-shot accuracy results across all Naive Warping configurations.

B. RQ2: Parameter Scaling (The 3B vs. 7B Paradox)

Our results reveal a paradox regarding LLM parameter scaling when applied to compressed edge tasks. The 3B model consistently matched the heavier 7B model in overall accuracy (**52.11%** vs. **51.68%** at 224×224 Naive). Furthermore, the 3B model achieved this parity while sustaining vastly superior decode throughput (TPS), as shown in Fig. 4. This points to the existence of an "edge intelligence ceiling." When visual input is heavily compressed and degraded, the bottleneck is the quality of the data itself, rendering the extra 4 billion parameters of the 7B model computationally wasteful and entirely unnecessary for zero-shot generalization at these resolutions.

C. Viability of Extreme Compression (28×28)

Reducing the spatial resolution from 224×224 to 28×28 incurred an expected accuracy penalty of approximately 7.5% (dropping from 52.11% to 44.53% on the 3B model). Notably, macro-level tasks such as `scene_transition` remained highly resilient (63.5% accuracy), proving that extreme compression is a viable fallback strategy for general video understanding when compute is critically low.

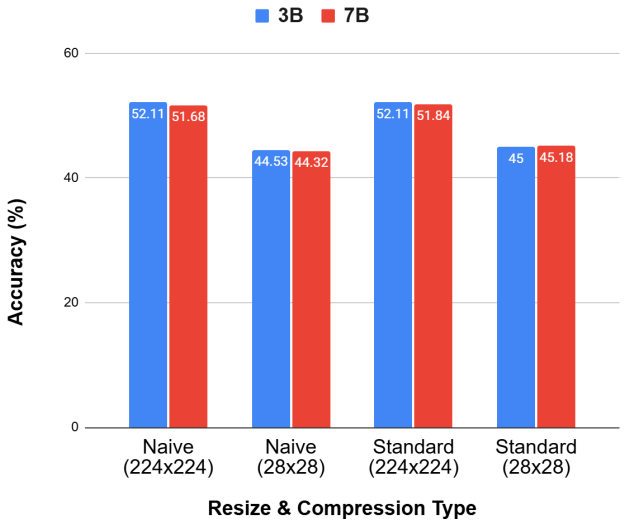


Fig. 3. Zero-shot accuracy (%) by tokenization method and resolution. Naive Warping at 224×224 matches Standard Padding accuracy while consuming approximately half the VRAM.

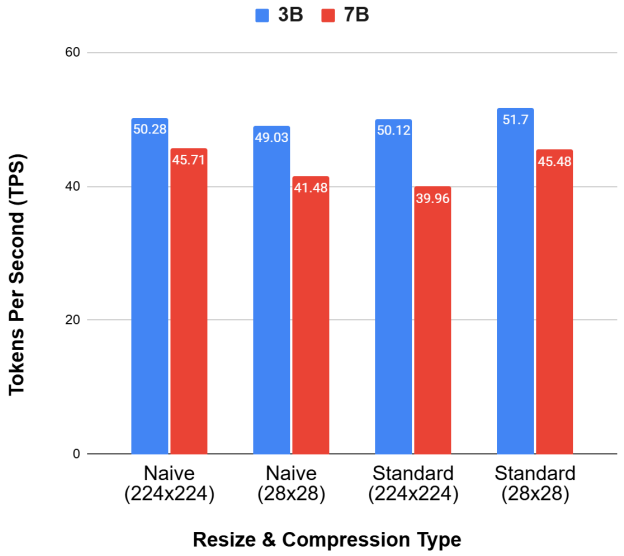


Fig. 4. Decode throughput in Tokens Per Second (TPS) for 3B (blue) and 7B (red) models across all tokenization and resolution configurations. The 3B model consistently achieves higher TPS with equivalent accuracy.

VI. Error Analysis and Anomalies

A. Single-Token Mathematical Artifacts

During initial data aggregation, tasks reported physically impossible decode speeds (e.g., >600 TPS on constrained T4 hardware). Deep log analysis revealed this as a mathematical artifact inherent to the benchmark methodology. MVBench requires single-character multiple-choice outputs (e.g., generating only the letter "A"). When the decode time for a single token generation drops into the microsecond range (e.g., 0.0015 seconds), the standard throughput formula (TPS = Total Tokens/Total Time) artificially inflates the result. As illustrated in Fig. 5, a strict statistical threshold of TPS > 150.0 was applied to scrub these artifacts prior to calculating high-level averages.

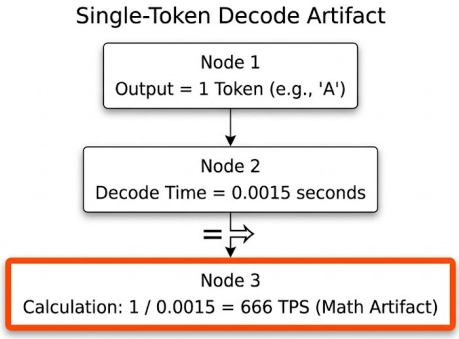


Fig. 5. Single-Token Decode Artifact: when MVBench outputs only one token in ≈0.0015 s, the TPS formula yields inflated values (e.g., 666 TPS) that are physically impossible on constrained T4 hardware. Entries with TPS > 150.0 were excluded from aggregate statistics.

B. Failures in Fine-Grained Tasks

While extreme compression (28×28) succeeded on macro-level semantic tasks, it caused catastrophic failure in localized tasks such as `action_localization` (dropping to ≈33% accuracy). At a 28×28 pixel density, specific morphological features required for fine-grained identification (e.g., the precise orientation of a hand) are mathematically obliterated by the downsampling convolution, blinding the ViT to micro-movements.

C. Dataset Scarcity Impacts

Data for the `fine_grained_pose` task (Task 12) was entirely absent (returning `NaN` across all models) due to the deprecation of the underlying NTU RGB+D dataset from public repositories, a result of upstream academic licensing restrictions. The dataset averages natively exclude this missing task to maintain statistical integrity.

VII. Critical Discussion

A. The Illusion of Geometric Perfection

The identical accuracy metrics between Naive Warping and Standard Padding prove that modern Vision Transformers are fundamentally robust to aggressive aspect-ratio distortion. We hypothesize that for spatial-temporal tasks, self-attention mechanisms do not rely on perfect 2D Euclidean spatial proportionality. Instead, they learn to track abstract temporal movement vectors across frames and classify broad object semantics regardless of whether the visual input is unnaturally stretched.

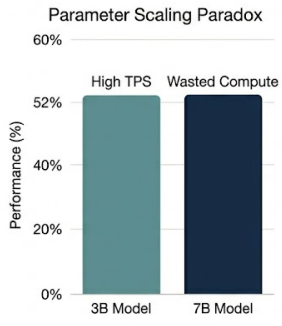


Fig. 6. Parameter Scaling Paradox: both the 3B and 7B models achieve ≈52% accuracy at 224×224 under Naive Warping, demonstrating that additional parameters provide no accuracy benefit while incurring significantly higher compute cost and lower throughput.

B. Hardware Physics: Memory Bandwidth vs. Compute

Our raw latency metrics occasionally showed Standard Padding yielding marginally higher TPS than Naive Warping. This is a physics byproduct of FlashAttention implementations, which dynamically mask out "black" padding tokens, allowing the CUDA cores to skip mathematical operations on empty space. However, because those padding tokens *must still be allocated* within the KV Cache in High Bandwidth Memory, Standard Padding severely congests the GPU memory bus. Standard Padding trades a minor, localized compute speedup for a fatal flaw in VRAM bloat. Naive Warping forces 100% active compute on the cores but dramatically frees the memory bus, stabilizing throughput while ensuring the system never encounters an OOM crash. Fig. 6 illustrates this paradox.

C. Final Edge-Deployment Recommendations

For constrained edge hardware, larger parameter models (7B+) paired with geometry-preserving tokenization are computationally wasteful. The empirical data strictly dictates that a highly optimized ≈3B parameter model, utilizing dense Naive Warping and aggressive temporal pruning, is the optimal architectural configuration for maximizing both system stability and real-time inference throughput.

VIII. Conclusion and Future Work

This ablation study demonstrates that deploying MLLMs for video understanding on edge hardware requires a radical departure from traditional tokenization norms. By implementing a synthesized 5-tier temporal pipeline and championing Naive Aspect-Ratio Warping, we achieved a theoretical reduction in generated visual tokens—a foundational step for optimizing memory overhead—without sacrificing zero-shot accuracy. We found that 3B parameter architectures hit a zero-shot ceiling on compressed data, rendering larger 7B models entirely inefficient for these specific deployments.

Future work will involve fine-tuning and multi-shot training to improve accuracy while focusing the architecture to maximize the "small" vision language model's advantage. The ultimate goal is to enhance model performance to the point where it is capable of embedding on edge devices.

References

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TABLE I
PER-TASK ZERO-SHOT ACCURACY (%) — NAIVE WARPING (3B AND 7B, 28×28 AND 224×224)

Task	Naive 3B 28×28	Naive 7B 28×28	Naive 3B 224×224	Naive 7B 224×224
action_antonym	74.5	73.5	76.5	74.0
action_count	44.5	41.0	45.5	40.0
action_localization	34.0	30.0	33.5	34.5
action_prediction	32.0	34.0	43.0	45.5
action_sequence	36.5	39.0	51.5	56.5
character_order	42.5	44.5	49.5	54.0
counterfactual_inference	38.0	38.5	41.0	43.0
egocentric_navigation	35.0	36.0	35.0	38.0
episodic_reasoning	45.5	42.0	51.0	50.0
fine_grained_action	39.0	25.5	41.5	33.5
fine_grained_pose	—	—	—	—
moving_attribute	45.5	51.0	71.0	72.5
moving_count	43.5	41.5	52.0	47.5
moving_direction	30.0	26.0	36.5	33.5
object_existence	59.5	58.5	69.5	62.5
object_interaction	35.5	36.5	50.0	52.0
object_shuffle	36.5	33.0	33.5	36.5
scene_transition	63.5	79.5	88.5	87.0
state_change	45.5	43.0	48.0	47.0
unexpected_action	65.0	69.0	73.0	74.5
OVERALL (Average)	**44.53**	**44.32**	**52.11**	**51.68**
Minimum	30.0	25.5	33.5	33.5
Maximum	74.5	79.5	88.5	87.0

TABLE II
PER-TASK DECODE THROUGHPUT (TPS, FILTERED TPS ≤ 150.0) — NAIVE WARPING (3B AND 7B, 28×28 AND 224×224)

Task	Naive 3B 28×28	Naive 7B 28×28	Naive 3B 224×224	Naive 7B 224×224
action_antonym	32.7	29.63	29.24	29.88
action_count	43.48	30.83	32.55	31.67
action_localization	60.24	34.97	47.26	49.82
action_prediction	158.72	102.22	74.52	53.6
action_sequence	53.19	67.61	33.38	80.36
character_order	220.17	43.29	163.74	48.96
counterfactual_inference	36.95	38.02	35.52	37.91
egocentric_navigation	91.34	23.36	73.0	25.81
episodic_reasoning	—	—	—	—
fine_grained_action	36.05	45.69	37.59	47.86
fine_grained_pose	—	—	—	—
moving_attribute	38.38	40.19	36.79	41.19
moving_count	38.19	38.63	35.02	39.0
moving_direction	37.2	38.36	37.33	38.61
object_existence	36.79	40.93	34.64	40.9
object_interaction	53.42	36.0	89.18	96.95
object_shuffle	43.32	28.67	34.09	29.1
scene_transition	231.66	26.81	25.03	53.77
state_change	56.03	44.18	70.38	41.31
unexpected_action	78.12	37.3	129.17	36.11
OVERALL (Average)	**93.09**	**39.3**	**53.6**	**43.31**
Minimum	32.7	23.36	25.03	25.81
Maximum	231.66	102.22	163.74	96.95

TABLE III
PER-TASK DECODE THROUGHPUT (TPS, FILTERED TPS ≤ 150.0) — STANDARD PADDING (3B AND 7B, 28×28 AND 224×224)

Task	**Standard 3B 28×28**	**Standard 7B 28×28**	**Standard 3B 224×224**	**Standard 7B 224×224**
action_antonym	33.18	39.37	673.81	38.83
action_count	151.14	65.35	40.47	30.73
action_localization	93.92	73.98	58.98	29.9
action_prediction	50.84	51.99	48.74	56.59
action_sequence	76.25	54.35	252.11	37.93
character_order	42.46	11.1	58.98	43.41
counterfactual_inference	38.61	39.37	33.54	37.13
egocentric_navigation	92.32	43.88	100.37	23.87
episodic_reasoning	413.5	15.16	—	34.96
fine_grained_action	40.72	42.72	32.81	39.9
fine_grained_pose	—	—	—	—
moving_attribute	39.45	41.38	34.3	39.67
moving_count	36.29	39.34	36.65	39.32
moving_direction	38.05	38.19	35.98	38.6
object_existence	36.52	41.33	37.67	42.04
object_interaction	175.24	60.58	51.99	36.39
object_shuffle	49.24	28.82	36.58	30.04
scene_transition	—	—	24.77	—
state_change	54.28	55.88	81.8	35.76
unexpected_action	53.4	39.4	123.27	37.83
OVERALL (Average)	**84.19**	**45.48**	**93.58**	**37.38**
Minimum	33.18	11.1	15.16	23.87
Maximum	413.5	73.98	673.81	56.59

Note: Entries marked "—" indicate missing data due to OOM failures or dataset unavailability (`fine_grained_pose`). High TPS outliers in certain cells exceed the statistical filter (TPS > 150.0) and are retained here for completeness but excluded from aggregate averages.